

Sprint Planning Report

Sprint: 2 **Date:** 22 Nov **Product:** Hospital Management

1. Sprint Goal:

- **Statement:** The goal of this sprint is to complete the planned features and improve the system's overall usability and efficiency.

2. Selected Product Backlog Items (with Acceptance Criteria):

- **User Story 1:**
- **Title:** ID and Insurance Company Validation
- **Description:** As a patient, I want to enter my ID number and insurance company name on the hospital's website so that I can check if my insurance is valid.
 - **Acceptance Criteria:**
 - The patient can enter their ID number and insurance company name.
 - The system performs validation of the entered data using an API.
 - The system notifies the patient whether their insurance is valid or invalid.
- **User Story 2:**
- **Title:** Insurance Response Status
- **Description:** As a patient, I want to receive a response from the insurance company through an interface that tells me whether my request is accepted or rejected, so I can know if my medical expenses will be covered.
 - **Acceptance Criteria:**
 - After entering the insurance details, the request is sent to the insurance company.
 - The response from the insurance company indicates whether the request is accepted or rejected.
 - The patient is notified about the status of their insurance claim (Accepted/Rejected).

- **User Story 3:**
- **Title:** Apply Insurance Discount to Bills
- **Description:** As a patient, I want a discount to be applied to the total amount when paying the bills, so that I can benefit from the insurance coverage.
 - **Acceptance Criteria:**
 - The system calculates the discount amount based on the insurance coverage.
 - The discount amount is displayed on the payment screen before final confirmation.
 - The final bill is updated to reflect the discount.
- **User Story 4:**
- **Title:** Lab Test Result Upload
- **Description:** As a lab technician, I want to upload test results to the system so that doctors can access lab data without delay.
 - **Acceptance Criteria:**
 - The lab technician can upload test results securely through the system.
 - Doctors can access the uploaded lab results in real-time.
 - The system provides confirmation once the results are uploaded.
- **User Story 5:**
- **Title:** Employee and Insurance Management for Hospital Managers
- **Description:** As a hospital manager, I want to be able to view all employee details and supported insurance companies in the hospital system so that I can ensure smooth operations.
 - **Acceptance Criteria:**
 - The hospital manager can access all employee details within the system.
 - The list of supported insurance companies is available for the manager.
 - The data is organized and easily searchable.

- **User Story 6:**
- **Title:** Generate Billing and Payment Reports
- **Description:** As an accountant administrator, I want to generate billing and payment reports so that I can track hospital revenue and patient payments.
 - **Acceptance Criteria:**
 - The system allows the accountant to generate billing reports.
 - Payment report generation is available and integrated into the hospital system.
 - Reports can be exported for record-keeping and analysis.

- **User Story 7:**
- **Title:** Lab Test Request and Result Viewing
- **Description:** As a doctor, I want to request lab tests and view results digitally so that I can make faster medical decisions.
 - **Acceptance Criteria:**
 - The doctor can request lab tests via the hospital system.
 - The test results are displayed clearly and are accessible in real-time.
 - All relevant test information is linked to the patient's profile.

- **User Story 8:**
- **Title:** Secure Online Bill Payment
- **Description:** As a patient, I want to pay medical bills online securely (via Visa, Mada, Apple Pay) so that I can complete my payments easily from home.
 - **Acceptance Criteria:**
 - The patient can select their preferred payment method (Visa, Mada, Apple Pay).
 - The payment process is secure and validated.
 - The system confirms the successful payment and provides a receipt.

- **User Story 9:**
- **Title:** View Prescription and Lab Results Online
- **Description:** As a patient, I want to view prescription and lab results online so that I can stay informed about my health status.
 - **Acceptance Criteria:**
 - The patient can access their prescription details and lab results through their online portal.
 - The information is presented clearly and in a readable format.
 - All lab results are linked to the patient's profile.
- **User Story 10:**
- **Title:** Contact Customer Service via Chat
- **Description:** As a patient, I want to contact customer service via chat so that I can get assistance with my medical queries or issues.
 - **Acceptance Criteria:**
 - The patient can initiate a chat session with customer service.
 - The chat service connects the patient to a support agent.
 - The patient receives timely assistance from customer service.
- **User Story 11:**
- **Title:** Send Promotional Offers via Email
- **Description:** As a marketing administrator, I want to send promotional offers to patients via email so that I can attract more customers.
 - **Acceptance Criteria:**
 - The system allows the marketing admin to create promotional email templates.
 - The system integrates with email sending services.
 - The marketing admin can track sent emails and responses.

- **3. Sprint Backlog Breakdown (Tasks):**
- **User Story 1 – ID and Insurance Company Validation :**
- **Task 1:** ID and Insurance provider input Screen - Assign to: Joud, 10 hours
- **Task 2:** Insurance validation API integration - Assigned to: Sumaiya - Estimated Effort: 12 hours

User Story 2 – Insurance Response Status :

- **Task 1:** Build response interface (accepted/rejected) - Assigned to: Roaa - Estimated Effort: 14 hours
- **Task 2:** Connect UI with insurance response API - Assigned to: Bushra - Estimated Effort: 16 hours

User Story 3 – Apply Insurance Discount to Bills :

- **Task 1:** Discount calculation logic - Assigned to: Sumaiya - Estimated Effort: 12 hours
- **Task 2:** Apply discount in payment screen - Assigned to: Joud - Estimated Effort: 10 hours

User Story 4 – Lab Test Result Upload:

- **Task 1:** Upload module (UI+file validation) - Assigned to: Bushra - Estimated Effort: 14 hours
- **Task 2:** Backend integration to store lab results - Assigned to: Sumaiya - Estimated Effort: 16 hours

User Story 5– Employee and Insurance Management for Hospital Managers:

- **Task 1:** Employee management screen - Assigned to: Roaa - Estimated Effort: 12 hours
- **Task 2:** Insurance company management module - Assigned to: Bushra - Estimated Effort: 14 hours

User Story 6 – Generate Billing and Payment Reports:

- **Task 1:** Billing report generator - Assigned to: Joud - Estimated Effort: 10 hours
- **Task 2:** Payment report and export options - Assigned to: Bushra - Estimated Effort: 12 hours

User Story 7 – Lab Test Request and Result Viewing:

- **Task 1:** Lab test request form - Assigned to: Roaa - Estimated Effort: 8 hours
- **Task 2:** Link test results to patient record - Assigned to: Sumaiya - Estimated Effort: 14 hours

User Story 8 – Secure Online Bill Payment:

- **Task 1:** Payment gateway integration- Assigned to: Sumaiya - Estimated Effort: 16 hours
- **Task 2:** Payment receipt confirmation - Assigned to: Joud - Estimated Effort: 10 hours

User Story 9 – View Prescription and Lab Results Online:

- **Task 1:** Prescription viewing page - Assigned to: Roaa - Estimated Effort: 8 hours
- **Task 2:** Link lab results to the patient portal - Assigned to: Bushra - Estimated Effort: 12 hours

User Story 10 – Contact Customer Service via Chat:

- **Task 1:** Chat UI - Assigned to: Joud - Estimated Effort: 10 hours
- **Task 2:** Backend chat service connection - Assigned to: Sumaiya - Estimated Effort: 14 hours

User Story 11 – Send Promotional Offers via Email:

- **Task 1:** Email template and UI - Assigned to: Bushra - Estimated Effort: 8 hours
- **Task 2:** Email sending service integration - Assigned to: Sumaiya - Estimated Effort: 12 hours

4. Sprint Capacity and Velocity:

- **Team Capacity:** $4 \times 4 \times 21 = 336$ hours
- **Estimated Velocity:** 264
- **Any Factors Affecting Capacity:** No holidays or absence during the sprint, Backend worked is concentrated on one member which may reduce effective velocity.

5. Risks and Impediments:

Identified Risks:

1. Possible delays in integrating the insurance validation API.
2. Backend tasks assigned mostly to a single team member, increasing the risk of bottlenecks.
3. Potential issues with the lab result upload module (file size, security, or validation problems).
4. Dependencies between UI and backend tasks may cause delays if one side is late.

Mitigation Strategies:

- Begin API testing and integration early in the sprint.
- Re-distribute tasks if heavy workload causes delays.
- Validate file upload requirements at the start of the sprint.
- Use Daily Scrum meetings to identify and remove blockers quickly.

- **6. Definition of "Done":**

- All Acceptance Criteria are fully implemented.
- Code is complete and passes unit, functional, and integration testing.
- UI and backend components are fully integrated and working as expected.
- The work has been peer-reviewed and approved.
- Documentation (if required) is completed.
- The feature has been merged into the main branch without conflicts.
- The item is ready to be demonstrated in the Sprint Review.